Matthew Chesnut

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GitHub Portfolio

Education

University of Maryland | College Park, MD
- B.S. Mechanical Engineering — Expected 2027
- Engineering Honors Program (EHP)

College of Southern Maryland | LaPlata, MD
- Engineering Coursework
- Dean's List (all semesters)

Experience

Project Engineer Intern | Pillar Construction | Alexandria, VA
May 2026 – September 2026
- Incoming project engineering intern using tools including OST & QBID for estimation & proposal bids. Learning exterior insulation and finish systems (EIFS), sub framing (Knight wall), plaster air barriers (STO PABs), & cement panels (Nichiha panels).
- Learning operations & project management, with field experience. Learning VDC/BIM & site laser scanning & processing the scan data. Finishing the internship with a final focus & capstone presentation.

Stem Tutor | College of Southern Maryland | LaPlata, MD **|** Prince Frederick, MD
January 2025 – August 2025
- Tutored students in math through calculus III, calculus-based physics, and engineering courses.
- Assisted students in understanding concepts and problem-solving strategies. Built strong connections with students to create a supportive learning environment. Adapted teaching methods to meet individual learning needs.

Auction Operations Team Lead | Bunting Auctions | Owings, MD
April 2022 – August 2025
- Supervised and trained a team of 4 photographers, ensuring efficiency and quality standards for auction listings.
- Assisted customers with pickups, inquiries, and consignor contracts, driving new sign-ups and maintaining strong client relationships.

Projects

- **3-Axis Robotic Arm:** Designed and fabricated a robotic arm utilizing stepper motors, embedded motion control, and inverse kinematics for repeatable precision positioning. Developed Python-based software with real-time 3D visualization and serial communication for interactive robotic manipulation.
- **Geothermal Heat Pump Cycle:** Designed & modeled a dual mode 12.3kW thermal system capable of high efficiency space conditioning (COPs: 6.805 winter / 4.504 summer) using R-454B refrigerant. Performed thermodynamic cycle analysis and component sizing for both summer and winter operating conditions.
- **GPS Speedometer:** Built an Arduino-based GPS device to display real-time speed, satellites, and coordinates, programmed in C with a 3D Printed case.

Achievements

- Eagle Scout, Engineering Honors Program

Technical Skills

CAD: SolidWorks, Onshape | **Programming:** Python, MATLAB, C/C++ | **Manufacturing:** MIG Welding, Machining, 3D Printing, Fabrication | **Electronics:** Circuit Design & Analysis, Arduino | **Engineering:** Embedded Systems, Data Analysis, Simulation